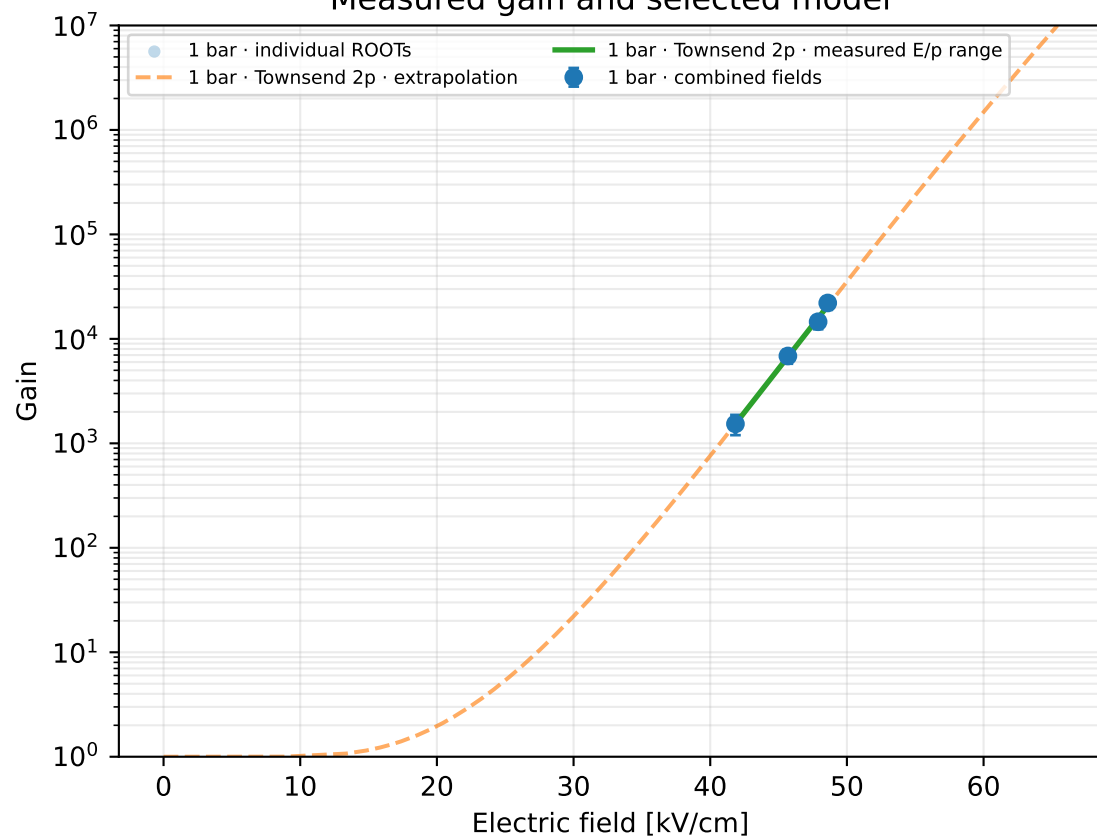
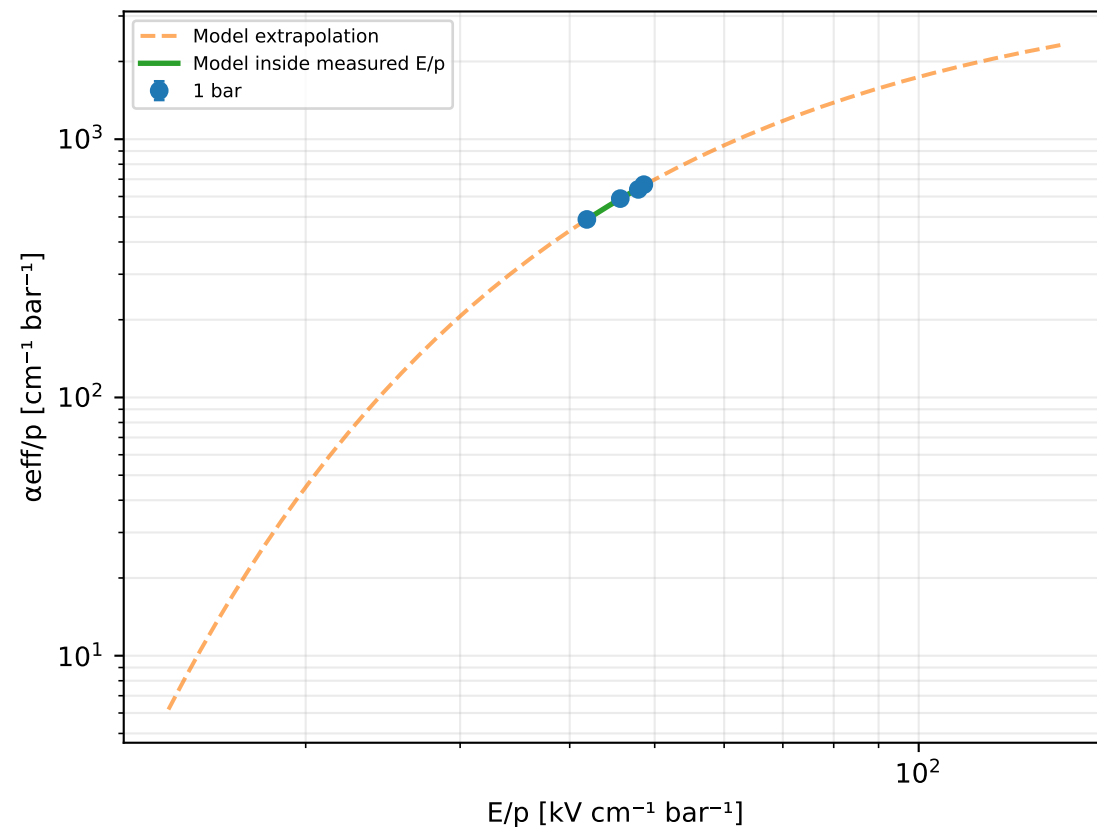


ArIsoCF4 · Ar 88 % / Iso 10 % / CF4 2 % · gap 0.15 mm

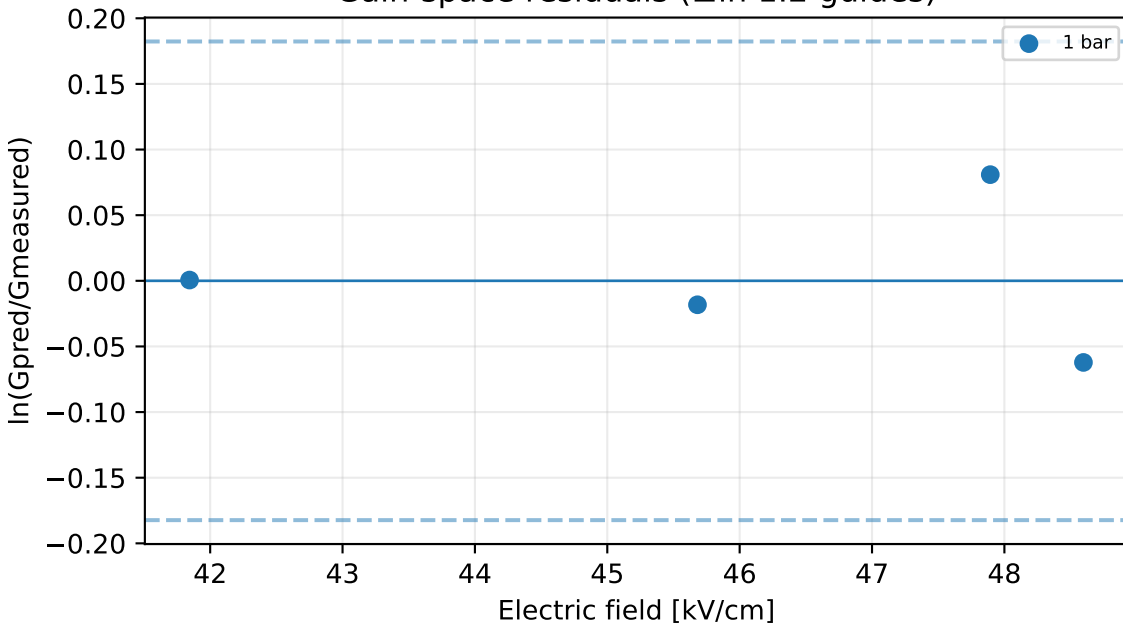
Measured gain and selected model



Reduced Townsend fit

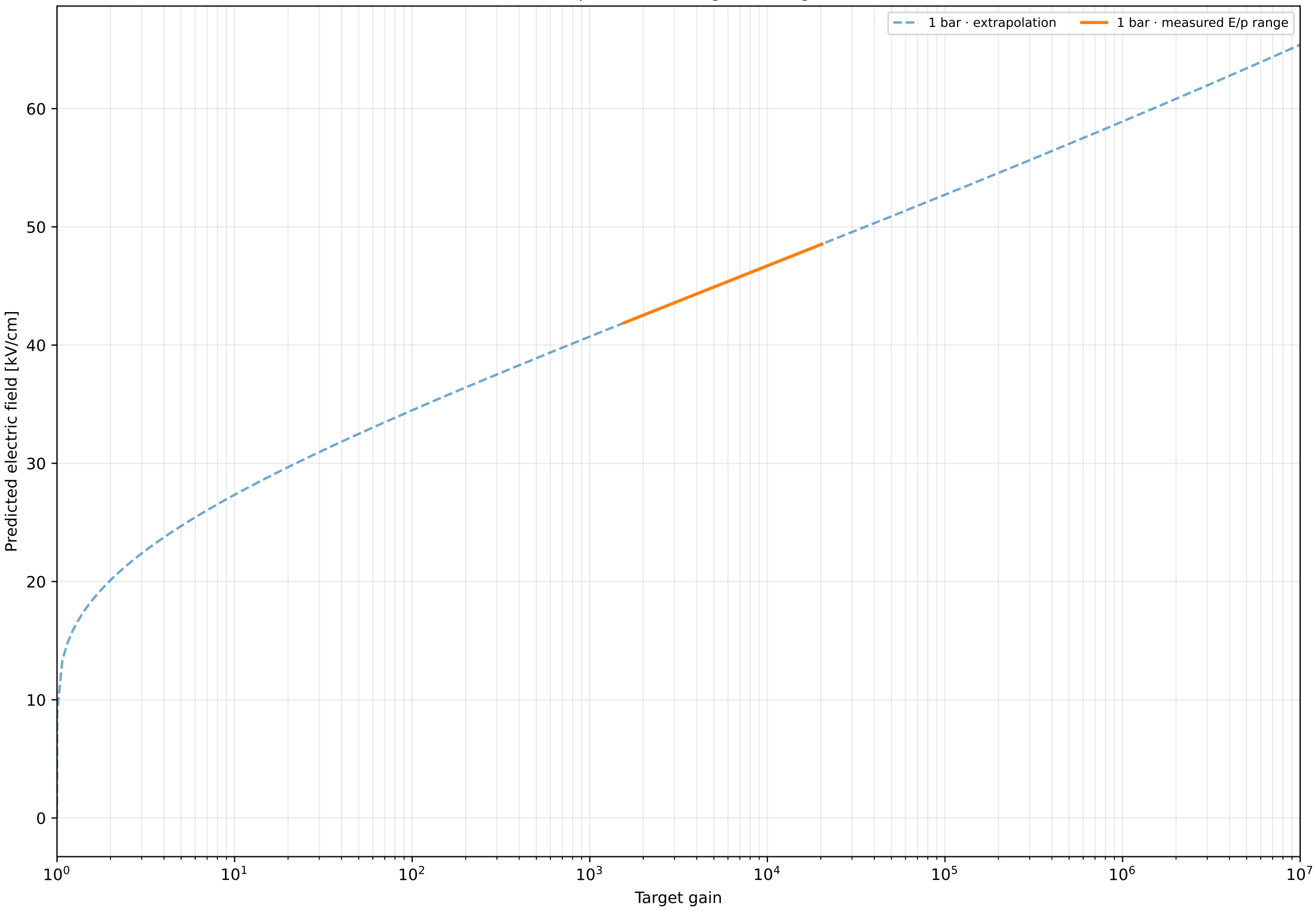


Gain-space residuals ($\pm \ln 1.2$ guides)



ROOT runs: 4
 Unique physical points: 4
 Pressures: 1 bar
 Selected model: Townsend 2p
 Status: rejected_poor_cross_validation
 Training RMSE: $\times 1.005$
 Cross-validation median: unavailable
 Cross-validation worst: unavailable
 Reduced-field range: 41.84–48.6 kV cm⁻¹ bar⁻¹
 A=4348, B=91.423, m=0, n=1
 Covariance condition: 1.75e+03
 Warnings: poor_cross_validation, large_cross_validation_outlier

Field extrapolation from gain 1 to gain 10^7



p [bar]	E [kV/cm]	E/p	Gain	σ Gain	α_{eff} [cm ⁻¹]	npe	runs
1	41.8443	41.8443	1535.05	338	489.088	20	1
1	45.6813	45.6813	6857.35	1.08e+03	588.872	20	1
1	47.8934	47.8934	14582.8	2.23e+03	639.173	20	1
1	48.5971	48.5971	22063.05	3.12e+03	666.777	20	1

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-25.9	—	—	no	poor cross validation, large cross validation outlier